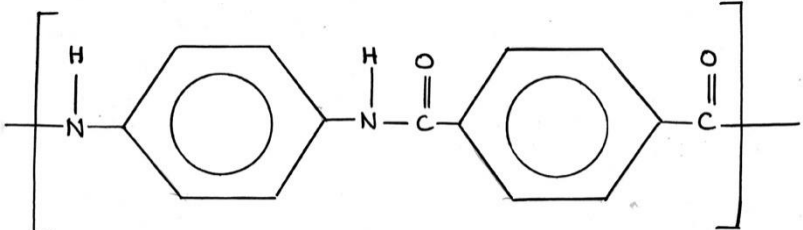


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
9.	(a)	(i)		hydrolysis	1					1
		(ii)		moles of NaOH = $\frac{75.00}{1000} \times 0.400 = 0.0300$ (1) total number of moles of acid is 0.0300 there are equal number of moles of each acid moles of R—COOH is 0.0150 $\Rightarrow$ same moles of R—COCl (1) M_r of R—COCl = $\frac{1.598}{0.0150} = 106.5$ (1)		2	1	2		3
		(iii)		award (1) for either of following M_r of the COCl group is 63.5 therefore M_r of the R group is 43 - likely to be C₃H₇ – the NMR peak area ratio 6:1 suggests (CH₃)₂CH formula of the acid chloride is (CH ₃) ₂ CHCOCl (1)			2			2
	(b)	(i)		HOOC-C ₆ H ₄ -COOH + 2PCl ₅ $\rightarrow$ ClOC-C ₆ H ₄ -COCl + 2POCl ₃ + 2HCl			1			1
		(ii)				1				1